

IoT-Based Smart Traffic Management System

Abstract:

Rapid urbanization and increasing vehicle ownership have made conventional fixed-time traffic signal systems inefficient, resulting in severe congestion, higher fuel consumption, increased carbon emissions, and delayed emergency response. This paper presents an IoT-based Smart Traffic Management System aimed at improving urban mobility through real-time monitoring and adaptive signal control. The proposed system employs infrared (IR) sensors, ultrasonic sensors, and intelligent CCTV cameras to continuously measure vehicle density and traffic flow at key intersections. Data collected from these devices are used to control traffic signal timings dynamically based on prevailing road conditions.

Additionally, the system integrates GPS-enabled priority mechanisms to provide uninterrupted green corridors for emergency vehicles such as ambulances and fire trucks. Experimental analysis demonstrates that the proposed framework can reduce average vehicle waiting time by up to 25%, decrease fuel consumption, and lower vehicular emissions. The architecture is scalable, cost-effective, and compatible with existing urban infrastructure, making it a practical solution for sustainable and intelligent Smart City traffic management systems.